Channel-Based Data-Subset Stability for Contrast Enhancement in Carotid Ultrasound

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ABSTRACT

Carotid ultrasound is widely used for the assessment of vascular anatomy and atherosclerotic disease; however, conventional delay-and-sum (DAS) beamforming is susceptible to clutter and incoherent echoes that reduce lumen contrast and obscure vessel-wall boundaries. This study presents a channel-based Data-Subset Stability (DSS) framework for adaptive contrast enhancement in carotid ultrasound by exploiting the stability of normalized correlations between delayed receive-channel signals across spatial lags. Raw radiofrequency data acquired with a 64-channel linear array were beamformed using conventional DAS while preserving the delayed receive-channel signals. Pixel-wise normalized inter-channel correlations were computed across multiple channel pairs and spatial lags, and their variation was used to generate a channel-stability weighting map. The resulting map was applied to the conventional DAS image to suppress unstable signal components while preserving coherent anatomical echoes. Preliminary qualitative results demonstrate suppression of residual lumen clutter and improved delineation of vessel-wall interface. The proposed framework provides a complementary adaptive weighting strategy for DAS and has the potential to improve visualization of vascular anatomy in carotid ultrasound.

Keywords: Carotid ultrasound; Adaptive beamforming; Data-subset stability; Channel stability; DAS; Image enhancement

1. INTRODUCTION

Carotid ultrasound is widely used for the noninvasive assessment of vascular anatomy, lumen morphology, intima-media complex, and atherosclerotic plaque. However, image quality can be degraded by clutter, noise, multipath reflections, off-axis scattering, and spatially varying speckle, reducing the contrast between anechoic regions, such as the vessel lumen, and surrounding hyperechoic structures [1, 2]. Such degradation may reduce diagnostic confidence during plaque characterization and vascular assessment.

Conventional scan-line delay-and-sum (DAS) beamforming is computationally efficient but does not explicitly distinguish coherent echoes from unwanted signal components. Consequently, adaptive approaches such as coherence weighting, and adaptive beamforming methods such as delay-multiply-and-sum (DMAS) [3, 4] and short-lag spatial coherence (SLSC) beamforming [5, 6] have been proposed to exploit inter-channel signal relationships for improving contrast and clutter suppression. In our previous studies, we introduced data stability methods, based on the principle that physically meaningful image components remain stable across different measurement configurations [7, 8], whereas artifacts exhibit greater variation. Here, this concept is extended to the receive-channel domain. Instead of generating multiple subset images, the proposed method evaluates the stability of normalized inter-channel correlations across multiple receive-channel pairs and spatial lags and uses this information to adaptively weight the conventional DAS image.

2. METHODOLOGY

Raw RF data acquired using a linear-array ultrasound system with 64 receive channels were reorganized into samples, receive channels, and scan lines and subsequently converted to complex IQ data. For each scan-line and axial depth

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position, the transmit and receive delays were calculated for each image pixel. These delays were then applied independently to the IQ data from each receive channel, resulting in 64 phase-corrected IQ signals at every image pixel, with each signal corresponding to one receive channel. The channel-wise delayed IQ signals were retained for subsequent channel-stability analysis. In parallel, the conventional full-aperture DAS image was obtained by coherently summing the delayed IQ signals from all 64 receive channels at each image pixel.

Channel-based DSS analysis was performed by evaluating the consistency of phase relationships among the receive channels. At each image pixel, the phase-corrected complex IQ signals from all 64 receive channels were available. For a given spatial lag (L), each channel (i) was paired with channel ($i + L$), and the complex conjugate product of the two signals was calculated. This was performed for adjacent channels ($L = 1$), i.e., [(1, 2), (2, 3), …, (63, 64)], and for channels separated by one additional element ($L = 2$), i.e., [(1, 3), (2, 4), …, (62, 64)]. Thus, lags 1 and 2 yielded 63 and 62 normalized measurements, respectively, providing a total of 125 measurements at each image pixel. The same approach can be extended to lags 1–5, resulting in 305 channel-pair measurements per pixel.

The variability of these complex measurements was then quantified at each image pixel using their coefficient of variation (CV), defined as the ratio of the standard deviation to the mean. This measure was used to characterize the consistency of the inter-channel similarity relationships. Coherent signals originating from anatomical structures are expected to produce similar relationships across receive channels and, consequently, low variability among the channel-pair measurements. In contrast, clutter and incoherent echoes are expected to exhibit greater variability. Therefore, a low CV indicates a stable and coherent channel response, whereas a high CV indicates an unstable response associated with clutter or artifacts.

The resulting CV map was spatially filtered and inverted to generate the channel-based DSS weighting map, such that regions with greater inter-channel consistency received higher weights. Finally, the DSS weighting map was multiplied by the envelope of the conventional full-aperture DAS image to enhance coherent anatomical signals while suppressing clutter and incoherent artifacts, producing the enhanced B-mode image. The overall processing workflow is illustrated in Figure 1.

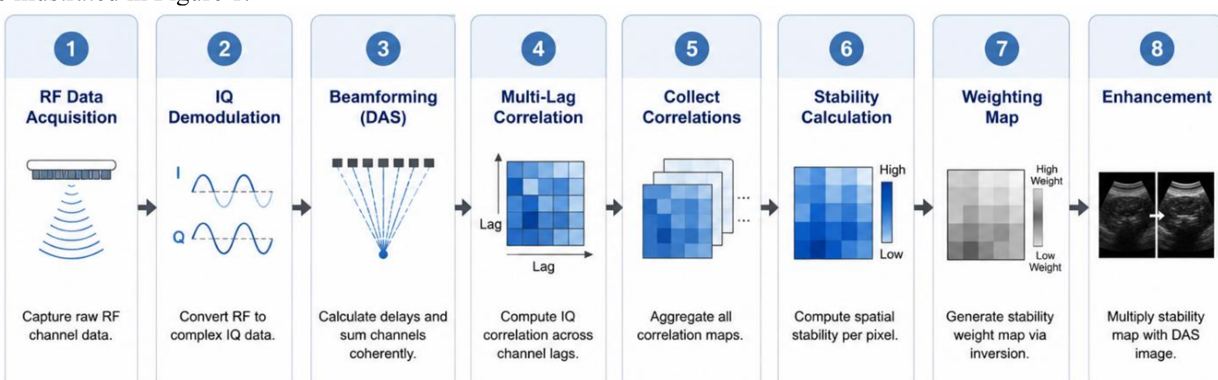

Figure 1: Workflow of the proposed Channel-Based DSS beamforming algorithm.

3. RESULTS

The proposed method was evaluated using carotid ultrasound data acquired from a 29-year-old male subject under an institutionally approved imaging protocol (No 41597-2). A Vermon 7–15 MHz linear-array transducer with 192 physical elements and a nominal center frequency of 9 MHz was used. The acquisition employed a 64-channel receive aperture, with a total of 191 scan lines. Representative results for lags 1 and 2 are presented in Figure 2, comparing conventional DAS beamforming, coherence-factor (CF), DMAS, SLSC with the proposed channel-based DSS processing.

Qualitatively, the proposed method improves image quality in both anechoic and hyperechoic regions. Notably, residual signals within the vascular lumen are substantially suppressed, corresponding to an approximate 15 dB improvement in clutter suppression within the carotid lumen. Concurrently, several vessel-wall interfaces and surrounding connective tissue structures exhibit enhanced conspicuity relative to adjacent tissues. These preliminary findings indicate that the channel-based DSS approach reinforces coherent anatomical features while attenuating unstable channel responses, without necessitating spatial smoothing. Figure 2 highlights representative regions where vessel-wall boundaries appear more distinct (yellow arrows) and where intraluminal clutter is reduced.

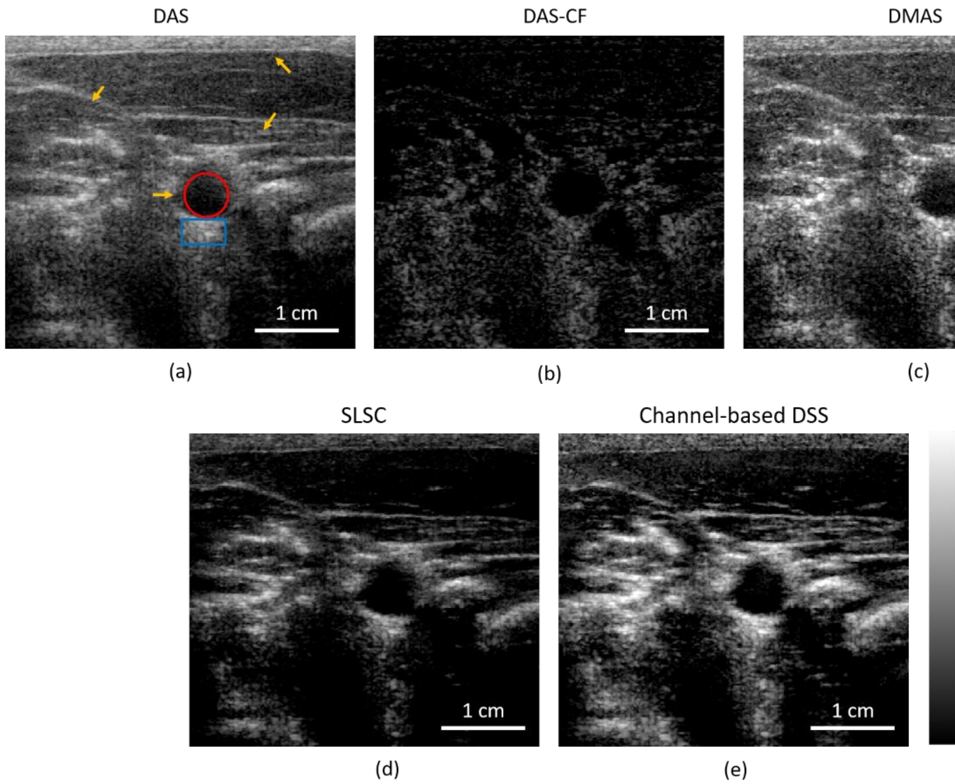

Figure 2. Representative carotid ultrasound images obtained using (a) conventional DAS, (b) DAS-CF, (c) DMAS, (d) SLSC and (e) the proposed Channel-Based DSS processing with lags 1 and 2. The proposed method reduces residual signals within anechoic regions while improving the conspicuity of hyperechoic structures. The dynamic range is 60 dB across (a, b, c, e) images.

Quantitative assessment was performed in accordance with [9], where the contrast ratio (CR) was computed using two regions of interest (ROIs): one within the lesion (indicated by the red circle in Figure 2) and the other in the surrounding background tissue (blue square in Figure 2). The resulting CR values for each reconstruction method are summarized in Table 1. Further quantitative analyses, including assessments of contrast, signal intensity, and clutter suppression, will be conducted for selected anechoic and hyperechoic regions to provide a comprehensive evaluation of the proposed method's performance.

Table 1: Quantitative contrast evaluation of the carotid artery using different methods.

Method	DAS	DAS-CF	DMAS	SLSC	**DSS (Proposed)**
CR (dB)	10.92	6.92	12.93	15.66	**16.57**

4. DISCUSSION AND CONCLUSION

The proposed weighting strategy is based on the hypothesis that coherent anatomical echoes remain stable across multiple receive-channel relationships, whereas clutter and incoherent scattering exhibit greater variability. By estimating this stability directly from delayed receive-channel signals, Channel-Based DSS generates an adaptive weighting map that complements conventional DAS beamforming without requiring iterative optimization.

Importantly, the proposed method introduces an estimated computational overhead of only ~ 0.1 relative to standard DAS, making it readily compatible with real-time, high-frame-rate vascular imaging. Furthermore, because stability is evaluated strictly within the spatial receive-channel domain for each individual transmission event (intra-pulse coherence), the method is inherently robust against pulsatile carotid wall dynamics and subject motion.

In conclusion, this work presents a Channel-Based DSS framework for real-time adaptive contrast enhancement in carotid ultrasound. Preliminary results demonstrate superior clutter suppression and enhanced vessel-wall delineation with minimal computational overhead. Future work will focus on extensive quantitative validation, optimal lag selection, and evaluation across larger clinical cohorts.

Disclosure of prior or concurrent submissions: This work is original and has not been previously published, presented, or concurrently submitted elsewhere.

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